

EML 3022: CAD Project

Fall 2025

Gustavo's Christmas Themed Wind-up Steam Locomotive Keychain

Gustavo Rodriguez

U7844-3708

Submission Date: 12-10-2025

Project Assembly Picture!

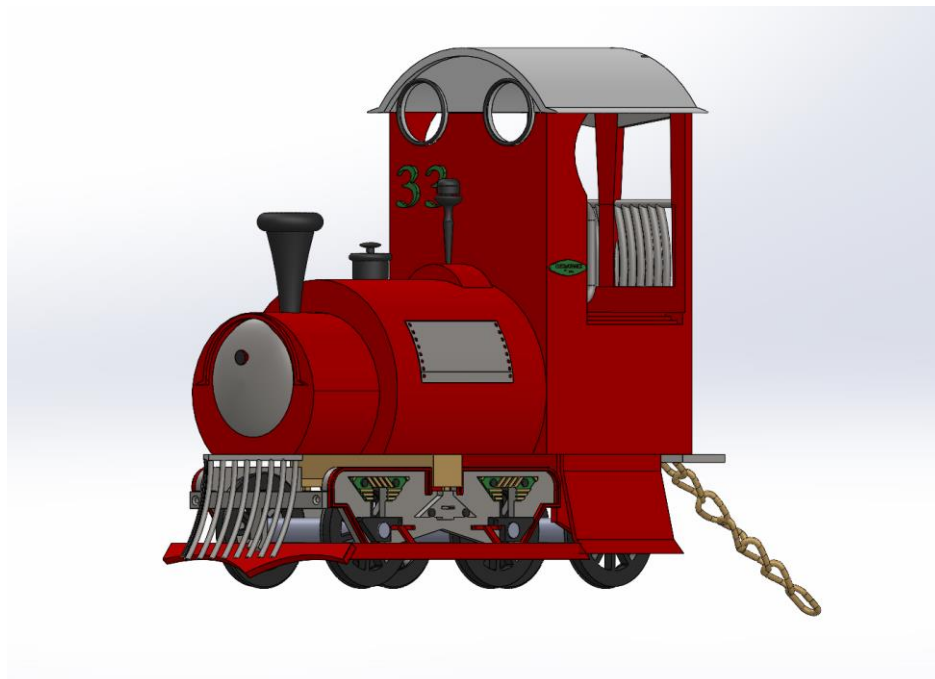


TABLE OF CONTENTS

Page

CAD Project Intro.....	2
I. Introduction to Design.....	3
II. Design.....	4
III. Assembly Drawings.....	5/6
5A-BOM.....	5
5B-Alternate/Back	6
IV. Part Drawings, 5C	7-13
V. Analysis	14
VI. Conclusion.....	15
VII. Appendix.....	16-17
Inspiration.....	16-17
VIII. SolidWorks Commands used.....	18

Introduction

For this project, I designed a steam locomotive keychain featuring a functional wind-up mechanism. The design was chosen because it could be Christmas themed for the season, and it could be scaled to ½ size for keychain use. This product could be sold as a set of randomized toys, such as in mystery boxes alongside other steam locomotive toys.

The external wind-up mechanism transfers rotational motion through a wheel axle attachment, allowing the model to propel itself, which allows for the future integration of designing small tracks or display platforms. I based my design on Saddle Tank locomotives, but added extra flairs such as the cowcatcher to increase appeal as this sort of cowcatcher wasn't typically present in these smaller saddle tank locomotives.

The design includes a license plate, some nameplates, Train number 33 plate, cab and cab roof, three sets of wheels and axles, the saddle tank and a firebox on the side, with side railings that have additional flair panels for ornamentation. The rear features a buffer and chain hook system, which were historically used to attach other cars but, in this design, will be used as a keychain to attach to a carabiner or keyring. This modernizes and lets the steam locomotive become a consumer project.

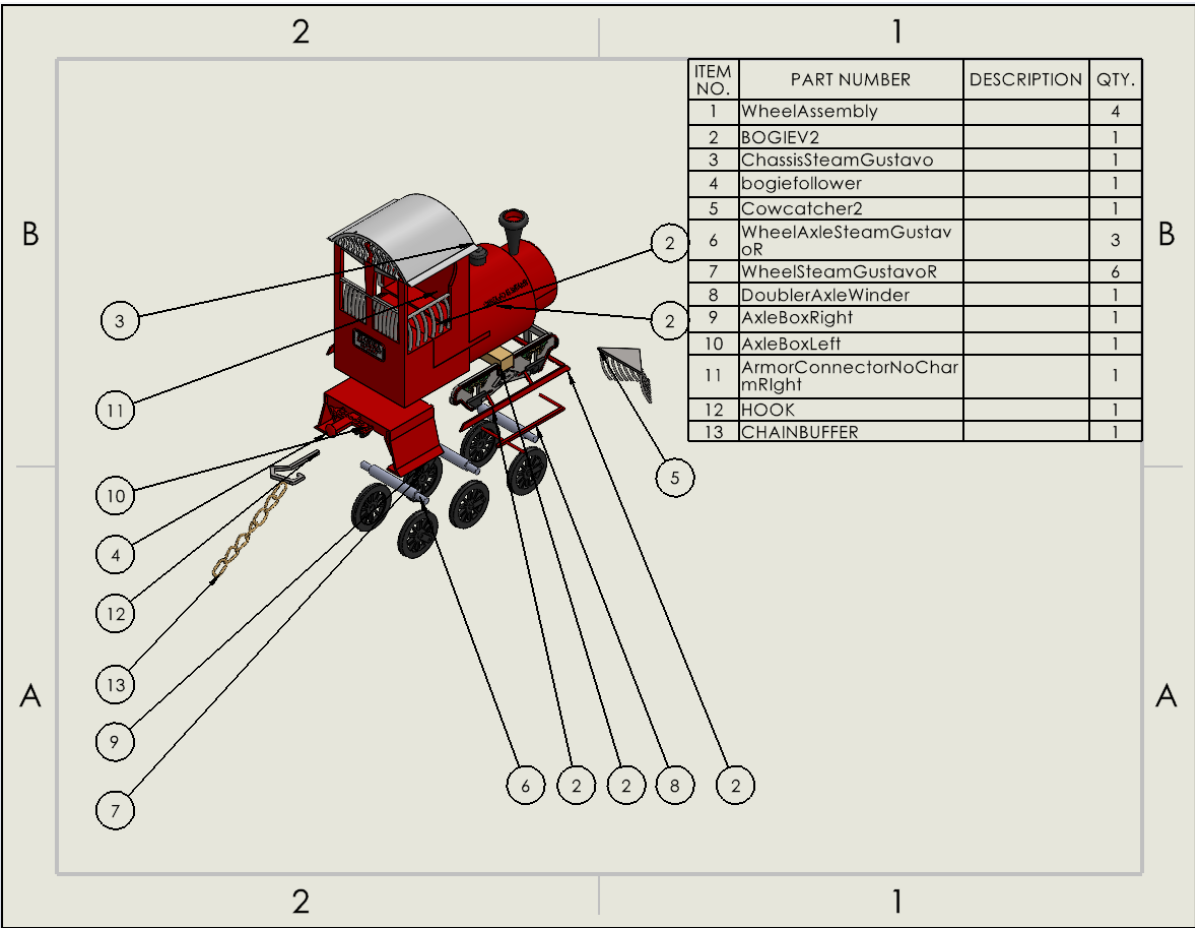
Design

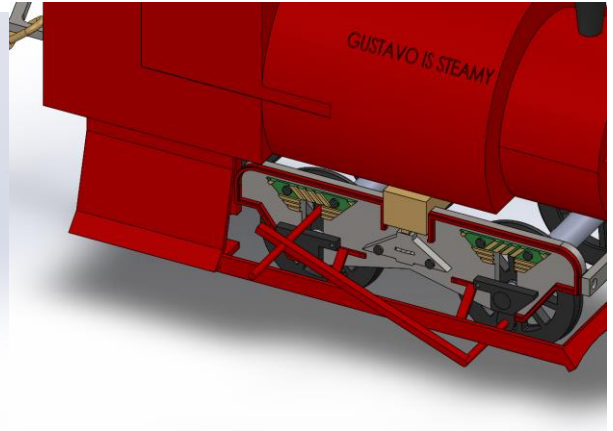
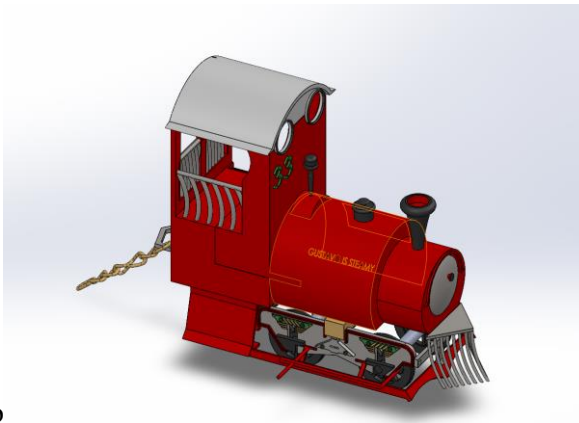
The main material for this design would be PLA, to allow for cheap and easy 3D printing. This would also allow for all the charms and details to print well, such as the name plates. This material will be ideal because it will be very light weight, which is what is needed when attached to the carabiner or keys. It will also allow for all the curved pieces to come out well given the printer is given the correct support. The PLA will also allow for the sharp cowcatcher to come out with its given design again with the correct support. Since the locomotive will not be carrying any load or stress, the material will be perfect to support the motions. The PLA is also perfect as colors appropriate to the Christmas theme can be used easily.

However, for the chain and hook I chose cast stainless steel, because of the way it looks as well as it's needed for proper attachment to the carabiner or key ring. The stress of the chain is the highest because it is carrying the entire design, so a material with higher versatility is needed.

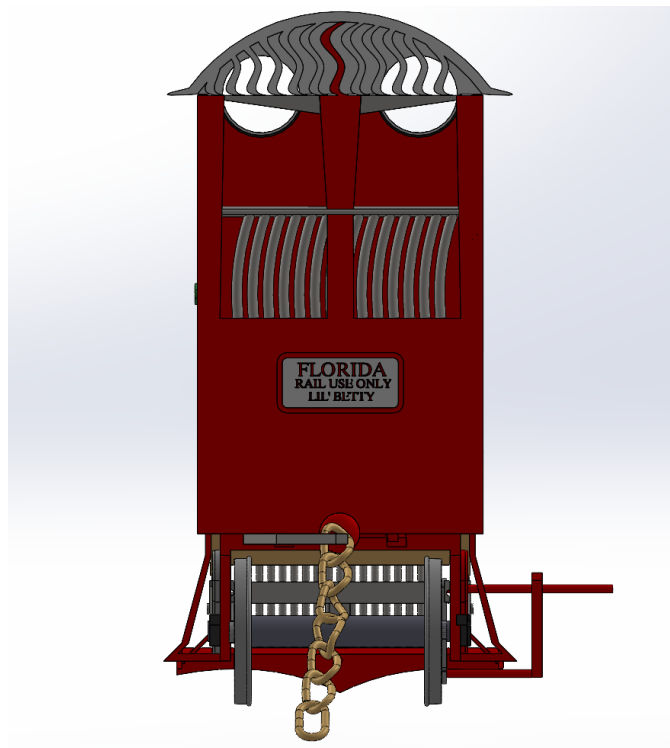
I faced difficulties in designing all the curvature; this model tested my usage of sweeps and revolve extrudes, due to the nature of the steam locomotives shape. I also encountered an issue in designing the chains, especially the links of them. Using only a linear pattern did not attach them; so, I had to discover the flex feature, which allowed for a 90-degree twist that attached the main body of them to each other.

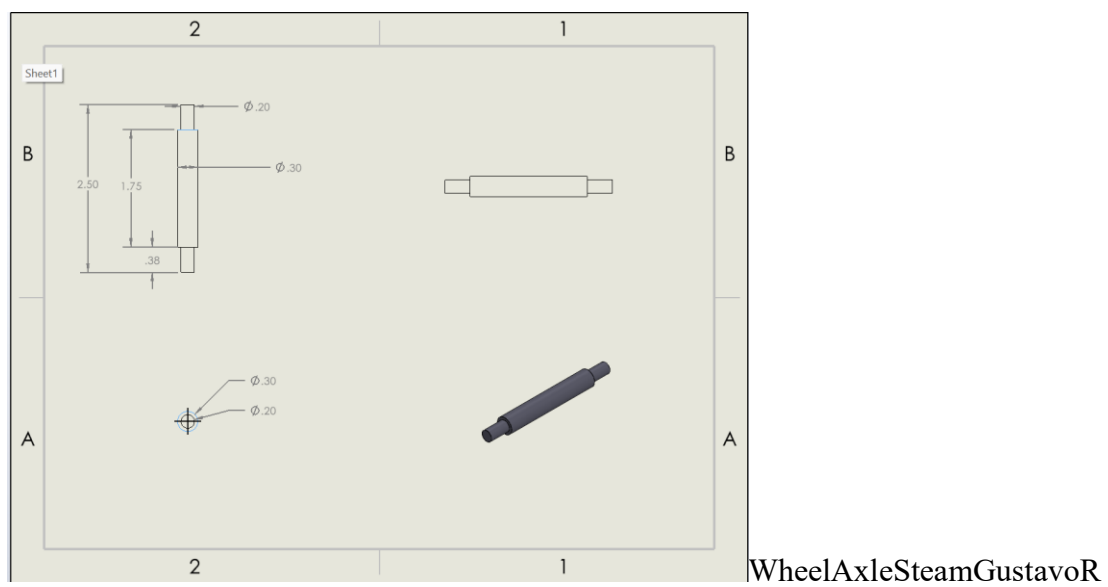
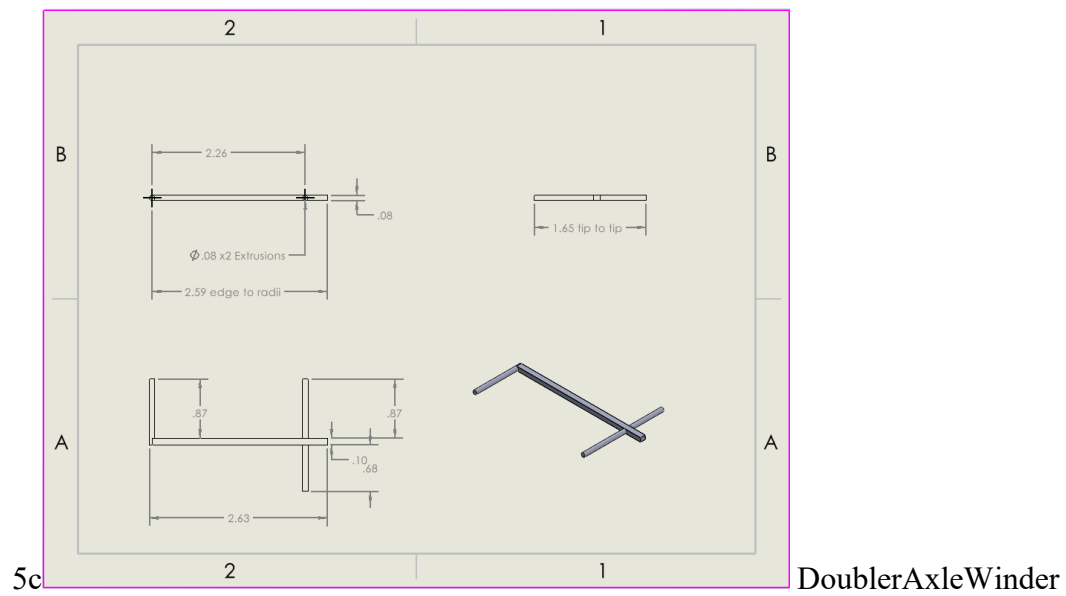
Assembly drawing in exploded view with balloons and BOM

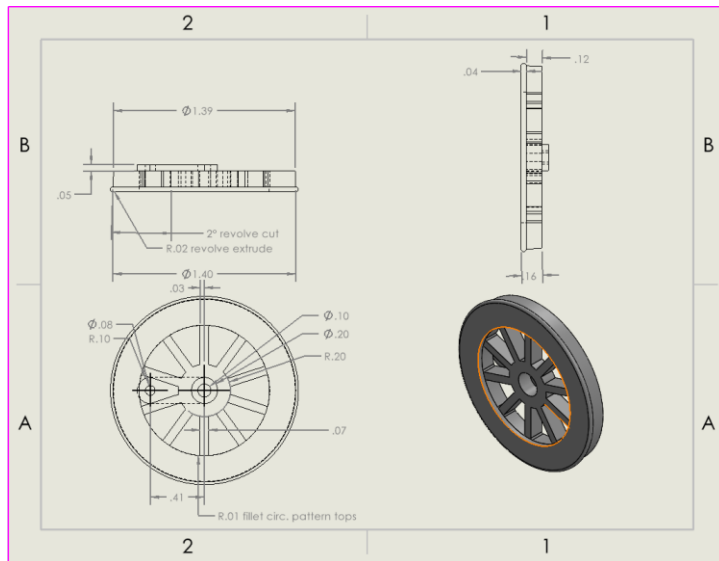




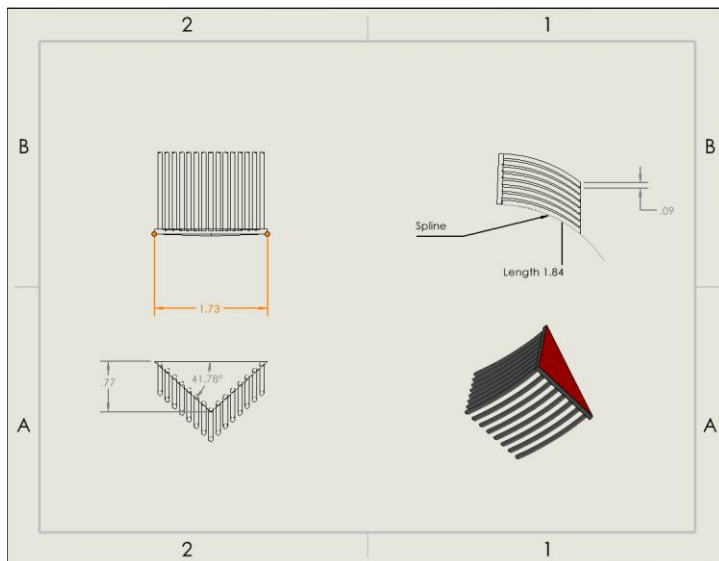
5b



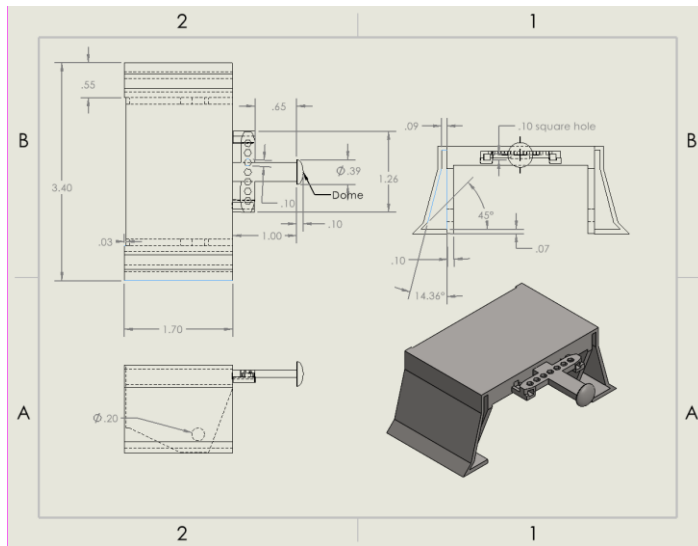




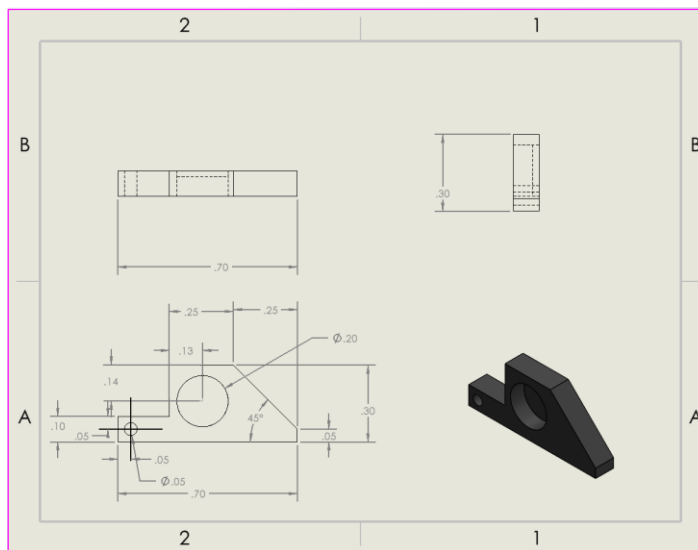
WheelSteamGustavoR



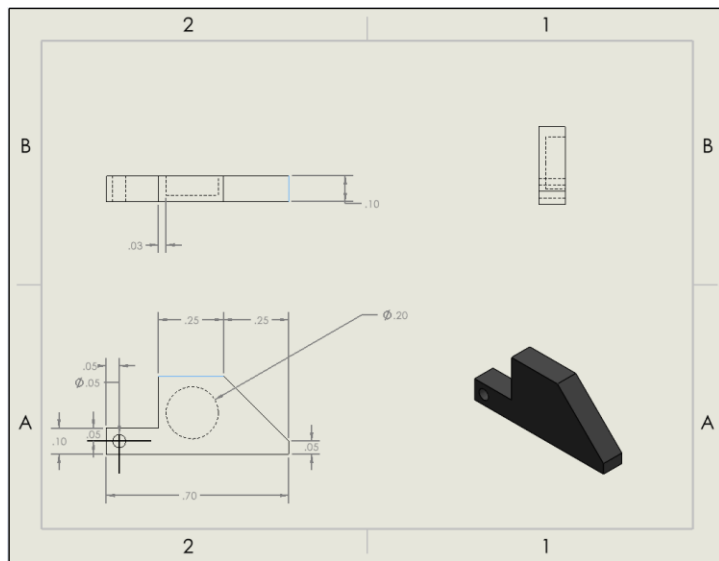
CowCatcher2



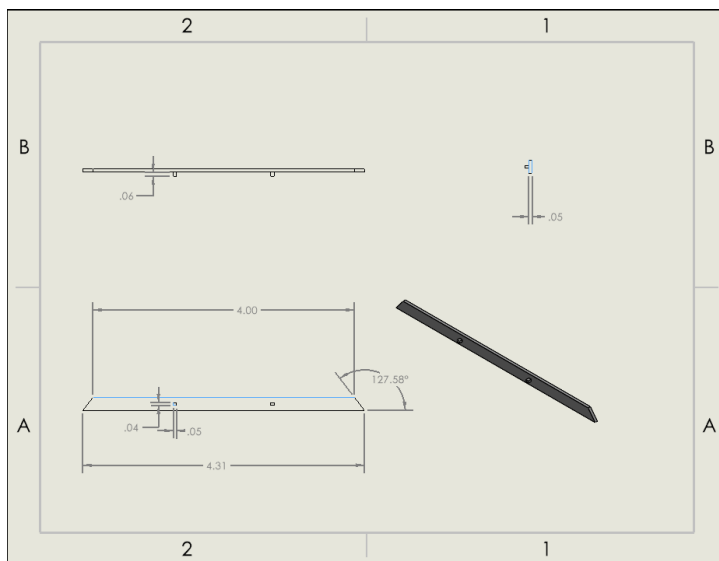
BogieFollower



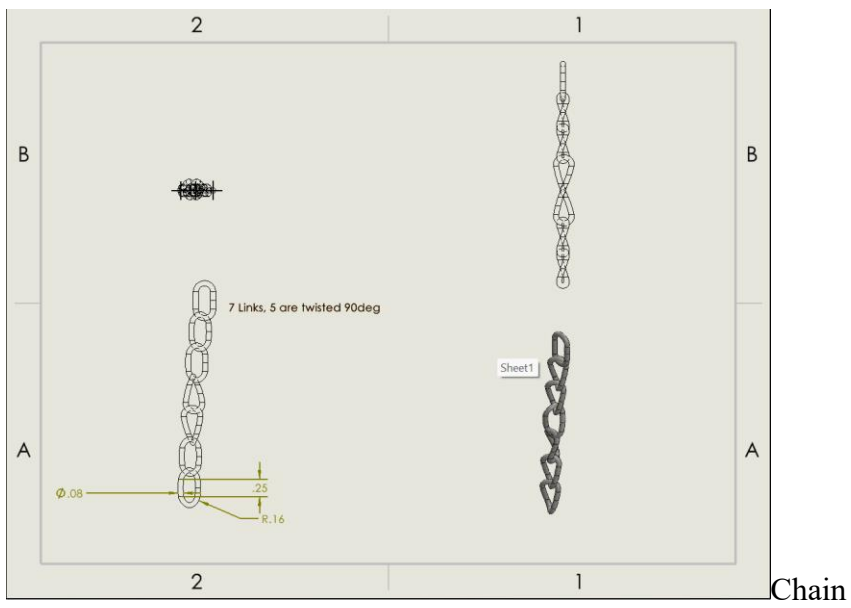
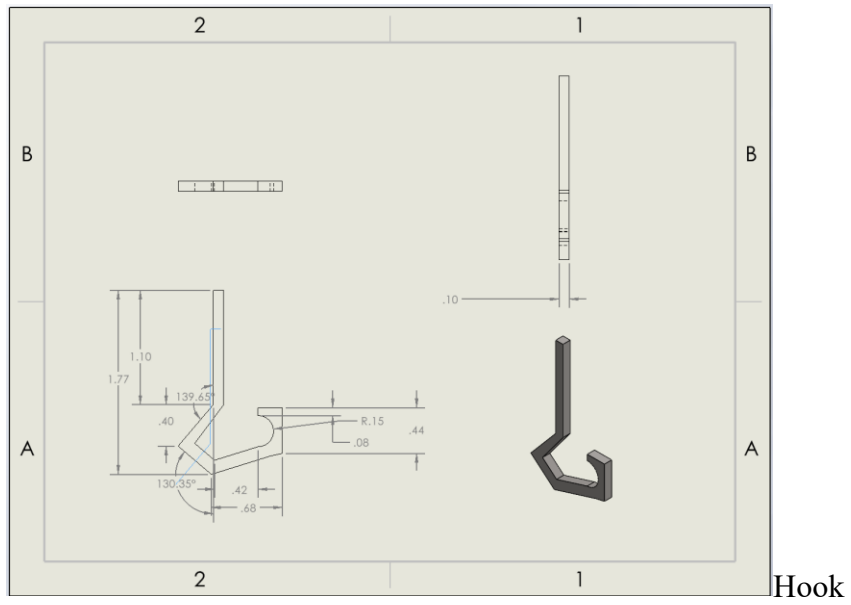
AxleBoxRight

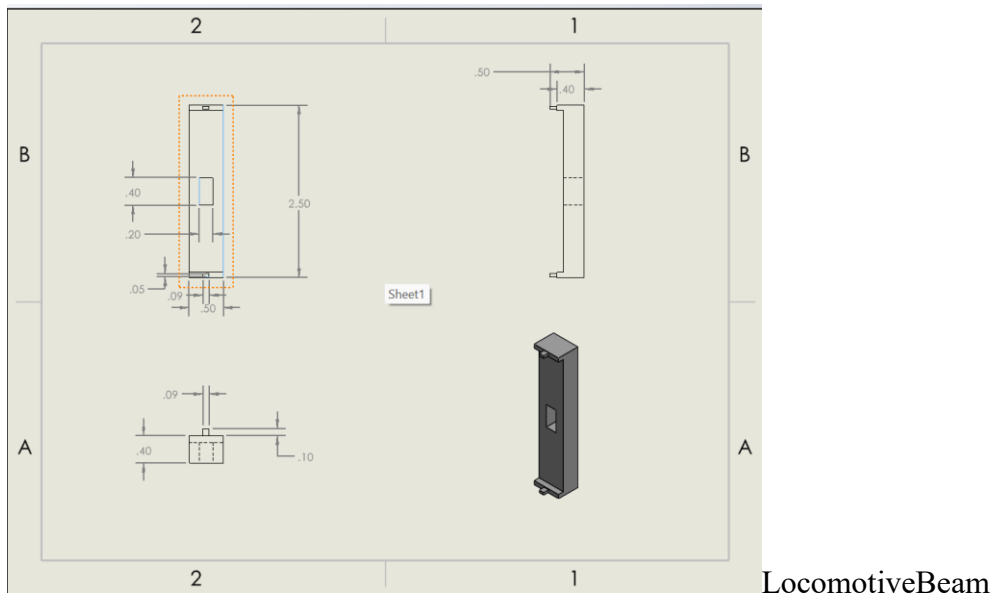


AxleBoxLeft

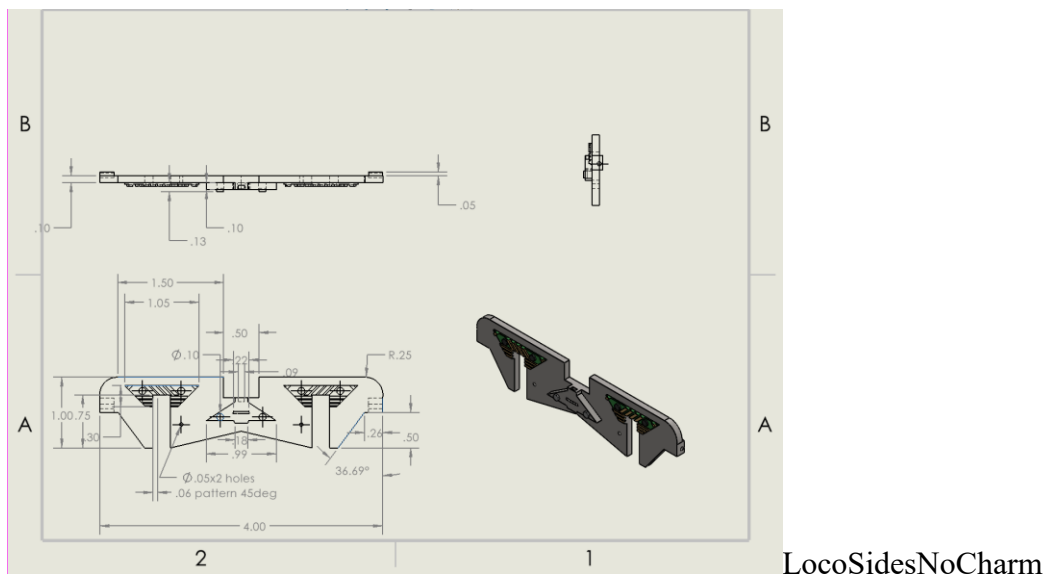


AxleBoxLeft

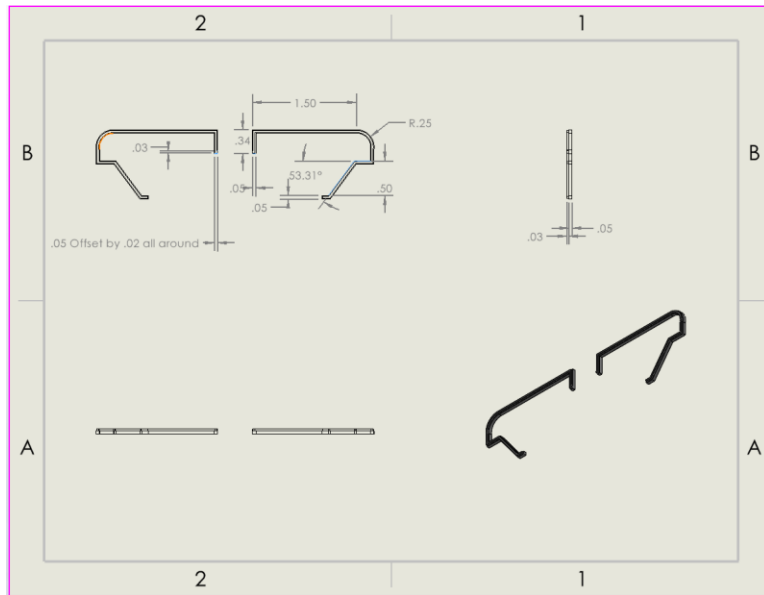




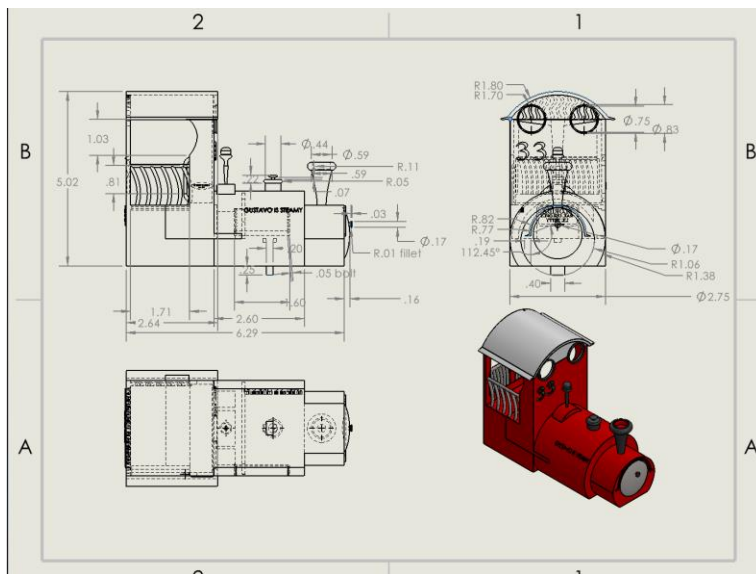
LocomotiveBeam



LocoSidesNoCharm



LocoSidesCharm



ChassisSteamGustavo

Analysis

The analysis demonstrated linear wheel motion with the 2nd axle following the 1st axles motion. The windup allowed the axles to move forward or backward. The animation confirms the assembly works in SolidWorks.

During the analysis all three of the wheel sets rotated smoothly in the same direction, due to the axle rotational motion demonstrating proper axle usage and spacing inside the Bogie (Bottom and wheel housing). The axles, axle box, and wheels were all concentric within the frame and did not interfere with the chassis.

However, the winder does interact with the sides of the locomotive, invalidating the right side of the motion. A path must be created for the winder. All other components are non-interfering and their geometry was modeled correctly.

This analysis doesn't reflect real life scenarios which could include friction. Given a track created to house the wheels, the locomotive should theoretically be able to ride on it.

Conclusion

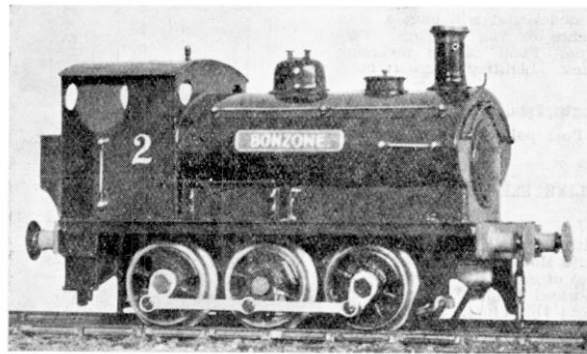
In the future I would add a better gear mechanism, to involve a brake system and to add a different side to the Locomotive that could house the rotation mechanism internally. As well as model more in the cab to include a steering wheel, and to design a track. Overall the Locomotive can move forward and backward based on the assembly, and has a beautiful Christmas theme that would be popular with all ages.

Appendix

Some Inspiration

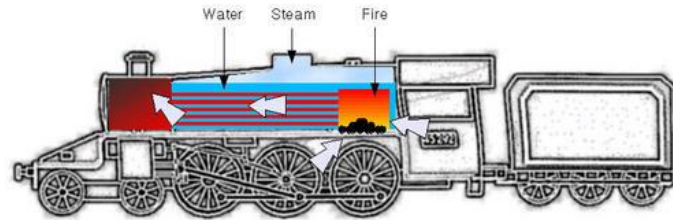


For the Chain/buffer hook



For Saddle Tank Design, this is the Kerr Stuart Tattoo 1921 and the Bonzone Peckett saddle-tank locomotive No2

https://www.brightontoymuseum.co.uk/index/Bonzone_Peckett_saddle-tank_locomotive_No2_%28Bond%27s_of_Euston_Road%29



For Firebox Placement I referenced this

<https://kevinboone.me/steam101.html>



I really enjoyed the Cowcatchers aggressive design on these Locomotives so I designed one similar.



I like how goofy this one's size is so I made my design Lil Betty(Nickname) be this size with similar features and name plates

Command Sheet

Features Checklist

A minimum of 10 different commands should be used from the following table

Put a checkmark next to the features that you used

Boss/Base	
Extrude..	✓
Revolve..	✓
Sweep..	✓
Loft..	
Boundary..	
Thicken..	
Cut	
Extrude..	✓
Revolve..	
Sweep..	
Loft..	
Boundary..	
Thicken..	
With Surface..	
Features	
Fillet/Round..	✓
Chamfer..	
Hole	
Simple..	
Wizard..	✓
Draft..	
Shell..	
Rib..	
Scale..	✓
Dome..	✓
Freeform..	
Deform..	
Indent..	
Flex..	✓

Pattern/Mirror	
Linear Pattern..	✓
Circular Pattern..	✓
Mirror..	✓
Curve	
Split Line..	
Projected..	
Composite..	
Curve Through XYZ Points..	
Curve Through Reference Points..	
Helix/Spiral..	
Reference Geometry	
Plane..	✓
Live Section Plane..	
Axis..	
Coordinate System..	
Point..	
Center of Mass..	
Mate Reference..	
Grid System..	
Tables	
Bill of Materials..	✓

Any other features used for your work

Curve Pattern

Estimate number of features
used in the project:

16

I used 15 different features,
but I estimate I used about 300 features total